

SOFTWARE ENGINEERING LAB-3

COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Student Startup Crowdfunding Platform

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1. Objective

To evaluate different architectural styles, select the most appropriate architecture for the Student Startup Crowdfunding Platform, and design a UML Component Diagram showing the major components, interfaces, dependencies, and interactions between them.

2. Scenario Overview

The system is a Student Startup Crowdfunding Platform that enables student founders to create and publish crowdfunding campaigns, while campaign backers can browse campaigns, select

reward tiers, and make pledges. The platform also supports milestone submission and review by a Faculty Committee before campaign funds are released. Secure payment processing is performed through a PCI-DSS-compliant payment gateway using tokenized payment authorization.

3. Key Requirements

Functional Requirements:

- Student Founders can create and publish crowdfunding campaigns.
- Campaign Backers can browse available campaigns.
- Campaign Backers can select reward tiers and make pledges.
- Student Founders can submit milestone evidence.
- Faculty Committee members can review and approve/reject milestone evidence.
- Approved campaign funds are released according to verified milestones.

Non-Functional Requirements:

- Payment transactions must use tokenized payment authorization through a PCI-DSS-compliant payment gateway.
- Campaign search and browsing results should load within 3 seconds under normal conditions.

4. Architectural Style Analysis

Architecture	Advantages	Disadvantages	Suitability
Layered Architecture	Clear separation of concerns, easy to understand and maintain	Some performance overhead and coupling between layers	High
Microservices Architecture	Independent scaling, fault isolation, independent deployment	Higher operational complexity, network latency, data consistency challenges	Medium
Client-Server Architecture	Centralized control, simple deployment and data consistency	Single point of failure and scalability bottleneck	Medium

5. Architecture Selection

Selected Architecture: Layered Architecture

We chose Layered Architecture for the Student Startup Crowdfunding Platform because the system has clear separation between presentation, business logic, data management, and external payment services. This architecture provides a simple and maintainable structure for implementing campaign management,

pledging, milestone verification, fund release, and secure payment processing.

Reasons for Selection

Reason 1 — Separation of Responsibilities

The platform contains different responsibilities such as campaign management, campaign discovery, pledges, milestone verification, fund release, and payment processing. Layered Architecture separates these responsibilities into logical layers, making the system easier to understand, test, and maintain.

Reason 2 — Suitable for the Current System

The crowdfunding platform follows a centralized application structure with a user interface, business operations, database, and external payment gateway. Layered Architecture fits this structure without introducing the additional operational complexity associated with independently deployed microservices.

6.Security Advantage

The Payment Service isolates payment-related operations from the rest of the application and communicates with a PCI-DSS-compliant payment gateway using tokenized payment authorization. This reduces exposure of sensitive payment information and provides a clear security boundary around payment processing.

7. Performance Benefit

The business and data layers can handle campaign search and browsing through efficient database queries and a dedicated data-access interface. This helps the system meet the requirement that

campaign search and browsing results should load within 3 seconds under normal operating conditions.

8. UML Component Diagram

The component diagram represents the major components of the Student Startup Crowdfunding Platform and the interfaces through which they communicate.

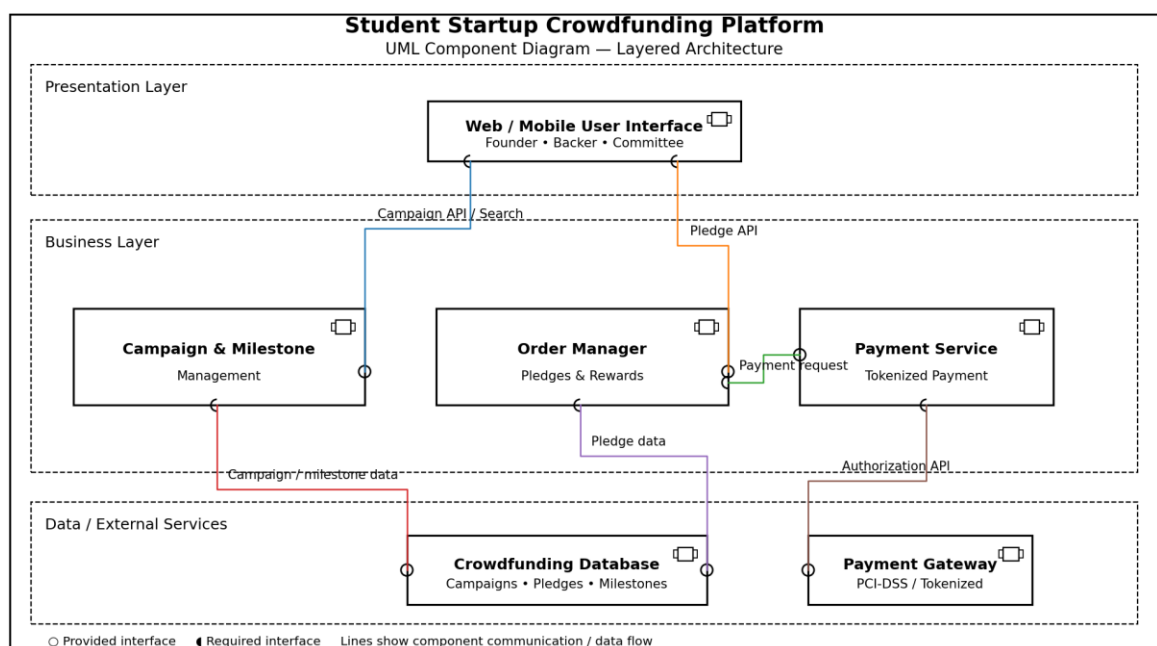


Figure 1: UML Component Diagram for Student Startup Crowdfunding Platform

9. Component Interfaces and Interactions

Interface	Components	Purpose
Campaign API / Search	User Interface ↔ Campaign &	Campaign creation, publication and campaign search

Interface	Components	Purpose
	Milestone Management	
Pledge API	User Interface ↔ Order Manager	Handles pledge and reward-tier operations
Payment Request	Order Manager ↔ Payment Service	Sends payment processing requests
Campaign / Milestone Data	Campaign Management ↔ Database	Stores campaign and milestone information
Pledge Data	Order Manager ↔ Database	Stores pledge information
Authorization API	Payment Service ↔ Payment Gateway	Performs secure payment authorization

10.Data flow

User



Web / Mobile UI



Campaign & Milestone Management



Crowdfunding Database

User



Web / Mobile UI



Order Manager



Payment Service



Payment Gateway

Order Manager



Crowdfunding Database

11.Conclusion

Layered Architecture is appropriate for the Student Startup Crowdfunding Platform because it provides clear separation between presentation, business logic, data management, and external services. It supports maintainability, secure payment processing, milestone-based fund management, and the required campaign browsing performance.